

**DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

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Year / Branch / Section: II / IT

Semester: IV

Academic Year: 2025-26



RAJALAKSHMI ENGINEERING
COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: Haripriya

Academic Year: 2025-26 Semester: IV Branch: IT

Register No

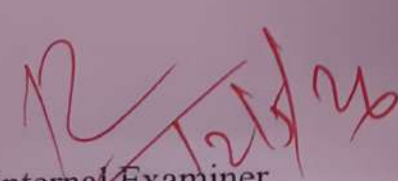
241001073

Certified that this is the bonafide record of work done by the above student in
the Software Construction CS23432 Laboratory

during the academic year 2025- 2026

Signature of Faculty in-charge

Submitted for the Practical Examination held on 12.5.26


Internal Examiner


External Examiner

INDEX

A1

NAME: E. Harshini

STD: IInd

SEC: 9th

ROLL NO: 0763

SUB: Software Construction

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		ONLINE BANKING SYSTEM		
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6)	1.04.26	Designing class & sequence diagrams for project Hierarchy		

7) 1.4.26 Designing Architectural
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8) 8.4.26 Testing- test plans and
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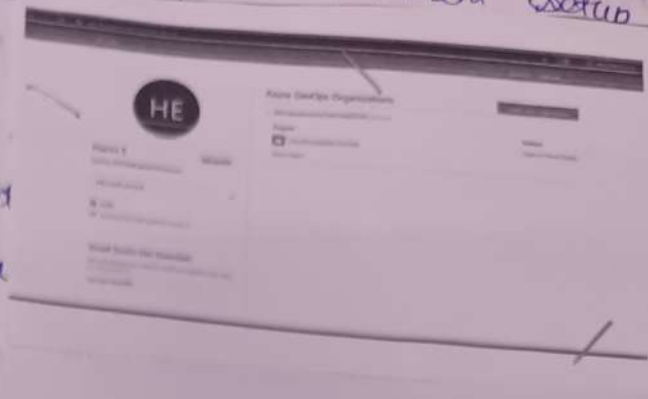
9) 15.4.26 Load testing and Pipelines

10) 22.4.26. GitHub : project
Structure & Naming
Conventions

Azure DevOps Environment Setup

Aim:

to setup
DevOps environment
through the Azure
Installation.



- 1) Open your web browser and go to the Azure site. <https://azure.microsoft.com/en-us/get-started/azure-portal> sign in using your Microsoft account, you can create one here <https://azure//signup?ref=msc>
- 2) Get started with Azure sign in Azure home page
- 3) Azure DevOps - Azure DevOps Organization
- 3) Open DevOps environment on the Azure platform by typing Azure DevOps organization in the search bar.
- 4) Click on the My Azure DevOps Organization link and create an Organization Home page.

Result:

Successfully created the Azure DevOps environment and created a new organization through the Azure portal.

Ex: 02

Azure DevOps project Setup and user story management

Aim:

To setup an Azure DevOps project for efficient collaboration and agile work management for an Online banking system

Procedure:

- 1) Create an Azure account
 - ⇒ Go to Azure DevOps and sign in with your credentials.
- 2) Create the first project in your organization
 - ⇒ Navigate to `dev.azure.com/[your-org-name]`
 - a) After the organization is setup, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more
 - b) On the organization home page, click on the New project button.
 - c) Enter the project name, description and visibility:

Name: online

Description: A

for customers to
and view transacti

a) Once you've
click create to set up your first project

3) Create New project details.

project name: Online banking system

visibility: public

version control: Git

work item process: Agile

click Create.

4) Project Dashboard

=> Once logged in, ensure you are in
the correct Organization.

=> click on the project name "online
banking system" and you should be taken
to the Azure DevOps project.

5) To manage user stories:

a) from the left-hand navigation
menu, click on boards. this will
take you to the main boards page.

where you can manage work items, backlogs and sprints.

b) On the work items page, you'll see the option to Add a work item at the top.

Azure DevOps Organisation



Result:

Successfully created an Azure DevOps project for the Online banking system. Then used story management and agile workflow setup.

Setting up epics, features and user stories for project planning.

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Procedure:

Create Epic, features, user stories, task

1) Fill in Epics:

Epic 1: Customer Account & Financial Transaction Management.

2) Fill in Features:

- => User Authentication & Profile
- => Fund transfer (Internal & External)
- => Bill payments
- => Transaction history & E-statements
- => Fixed Deposit Management

3) Fill in User story details:

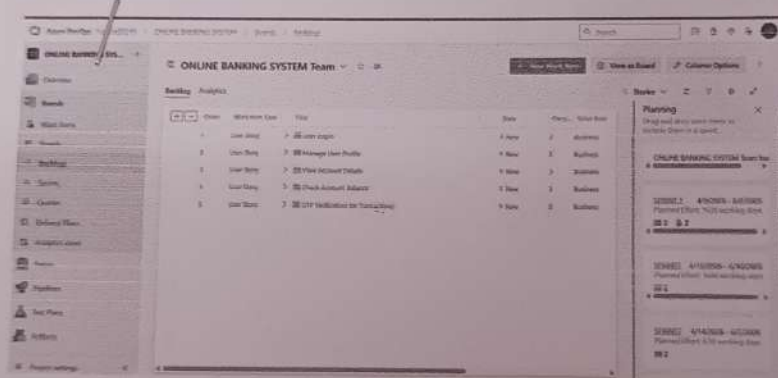
- => Account Balance check:

As a customer, I want to view my account balance on the dashboard so that I can monitor my funds easily.

As a user, I want to transfer money to another account using an IFSC code so that I can complete payments instantly.

⇒ Transaction security (OTP):

As a customer, I want to receive an OTP for every transaction so that my account remains secure for unauthorized access.



User
Story

Result:

Thus, the creation of epics, features, user story and task for online banking system has been created successfully.

Ex: 04:

Print planning

Aim:

To assign user story to specific print for the Online banking system

Print planning:

Print 1:

Taskboard

1. User Authentication & login:

⇒ Assigned to: Harini

Iteration 1:

Taskboard

2) Account Summary & Balance Inquiry;

⇒ Assigned to: Harish

Iteration 2:

Taskboard

3) Internal fund transfer:

⇒ Assigned to: Haripriya

Iteration 3:

4) Bill payment Integration:

⇒ Assigned to: Harini

5) Transaction Security & OTP:

⇒ Assigned to: Harish & Haripriya

Result:

The prints are created for the online banking system.

Ex: 05

Poker Estimation

Aim:

To create poker estimation for the user stories - Online banking system

Poker Estimation

Story points measure:

⇒ Effort

⇒ Complexity

⇒ Time

This uses fibonacci scale: 1, 2, 3, 5, 8, 13.
Story points for online banking system.

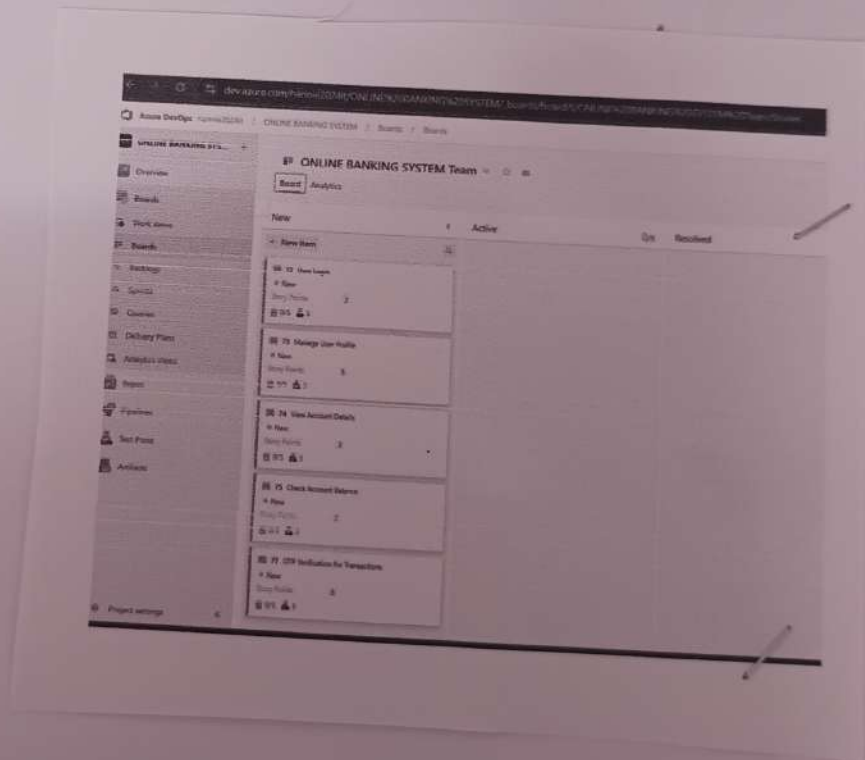
User story	Story points	Reasons
User Authentication & login	2	Secure login + session management
Account summary & Balance Enquiry	2	Simple data retrieval
Internal fund transfer	5	transaction logic + database update
Bill payment Integration	8	All integration + payment gateway
Transaction Security (OTP)	3	SMS API + verification logic

Total = 81 story points

Boards → Work Items → User Story → Story

Points → (1/2/3/5/8) → Same

Report for all user stories



Result:

The Estimation / story points is created for the project using poker Estimation.

Ex: 06

Designing class and sequence diagrams for project Arguments

Q/m:

To design a class diagram and sequence diagram for the given project

GA. class Diagram .

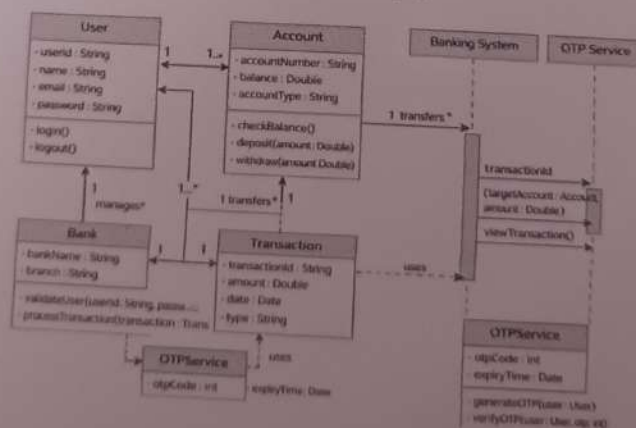
CLASS DIAGRAM

Class Diagram – Online Banking System

● Harsh E 10 Apr

Unfollow Edit

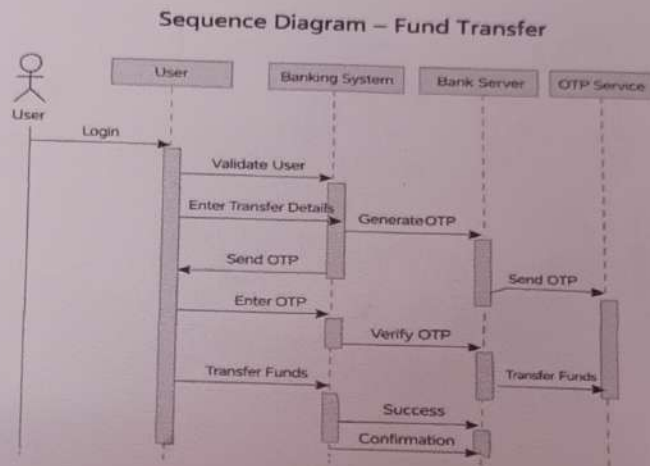
Class Diagram – Online Banking System



Sequence

Diagram

Sequence Diagram - Fund Transfer



Result:

The class diagram and sequence diagram designed successfully for the ~~new~~ online banking system.

Ex: 01

Designing Architectural and diagrams for project structure

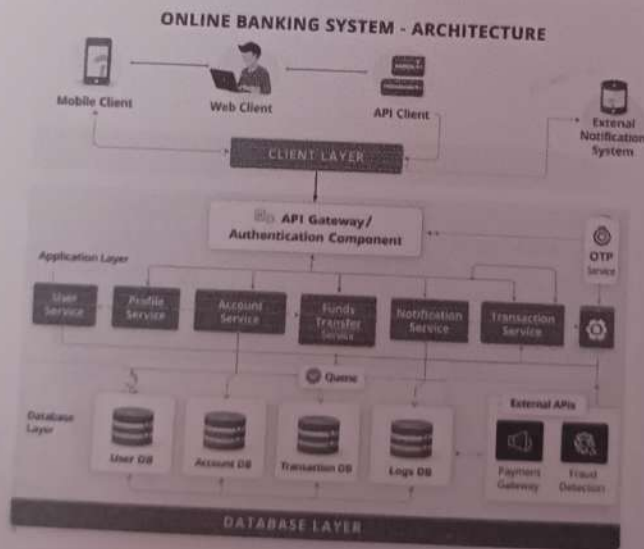
Aim:

To design and Architectural diagram and ER diagram for the given project.
QA: Architectural Diagram.

Architectural Diagram

ARCHITECTURAL DIAGRAM

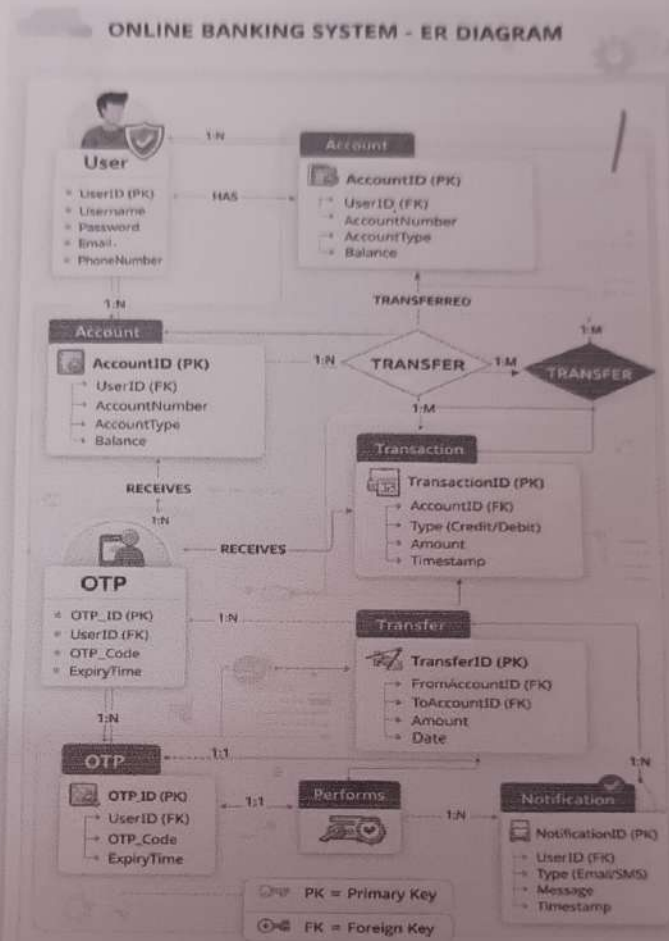
© 2012, 10/10/12



nd
structure

Diagram
project.

ER - Diagram



Result:

The Architecture Diagram and ER diagram is designed successfully for the online banking system.

Ex: 08.

Testing - test plans and test cases

Aim:

Test plans and test cases and write test cases for at least five user stories showcasing the happy path and error scenarios on azure devops platform.

Test planning and test case
Test case design procedure

Test execution is the process of running test cases to verify whether the application works as expected. Bug tracking is used to identify, record, and manage defects found during testing.

Azure devops provides tools for managing test plans, executing test cases, and tracking bugs efficiently.

PROCEDURE

STEP 1: Create test plan

login to Azure devops

Navigate to test plans.

Create a New Test plan for Online Banking System

STEP 2: Create test Suites.

Add test Suites Based on user stories:

User login

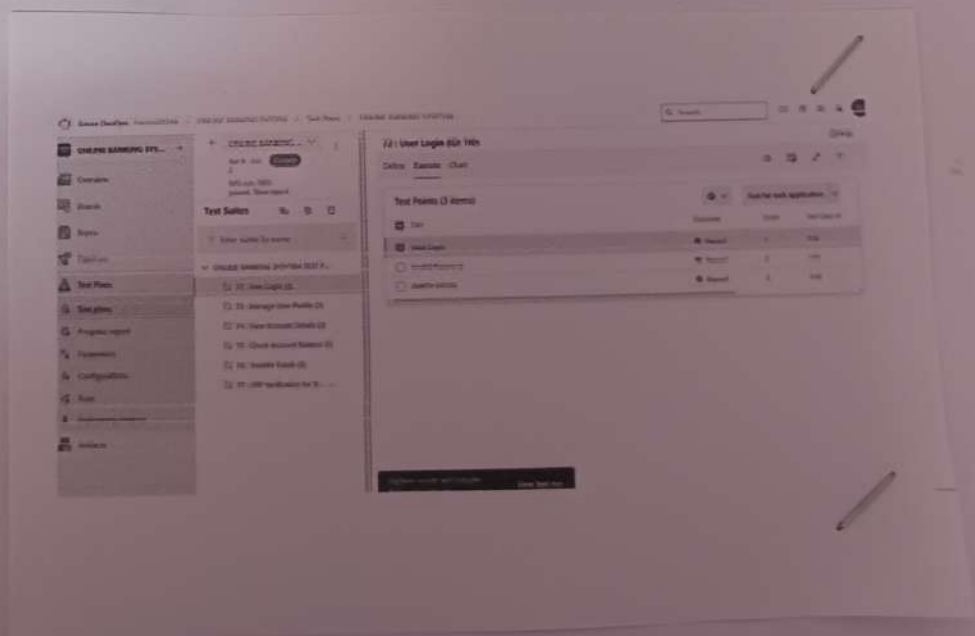
Manage Profile

View Account

check Balance

Transfer funds

OTP Verification.



STEP 3: Create test Cases.

Add Test Cases for Each Module.

Define steps and expected Results.

Step 4 :- Execute test Cases.

Go to Execute tab.

Run all test Cases.

Mark results as pass or fail.



STEP 5 :- Create Bugs

For failed Test Cases, click on 'Create Bug'.

Enter details :

Title

Description.

Steps to reproduce

Expected Result

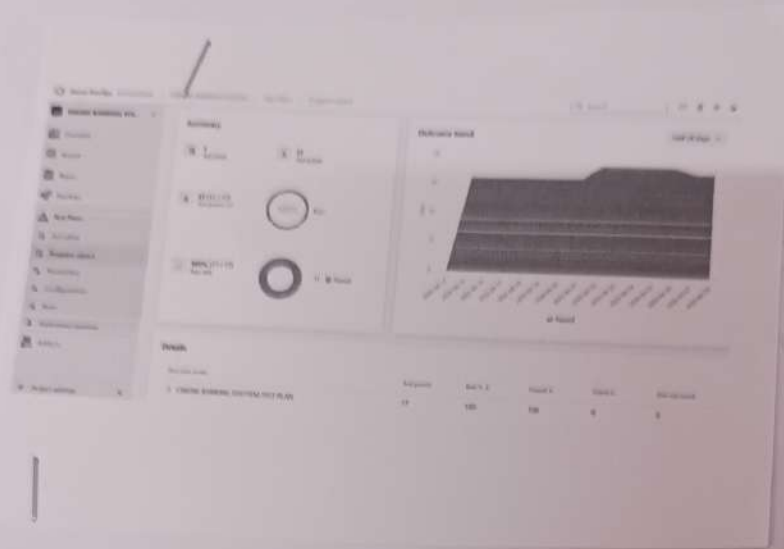
Actual Result

step 6 :- Update Bug status.

Change Bug state from New → Active → closed

STEP 7:- Analyze Results

View charts tab.
check pass / fail percentage



Result :-

Test cases were successfully executed for the Online Banking System.

Defects were identified and recorded as Bugs in Azure DevOps.

Aim:

To create an Azure load testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application build, test and deployment.

⇒ Load testing Setup:

The load test was executed in Azure load testing and metrics such as virtual users and response time were observed during execution

1) Creating Azure load testing Resource:

⇒ logged in to Azure portal

⇒ Create a new Azure load testing resource.

2) Configure load test

⇒ entered test name and description

3) load settings

⇒ Simulated multiple virtual users

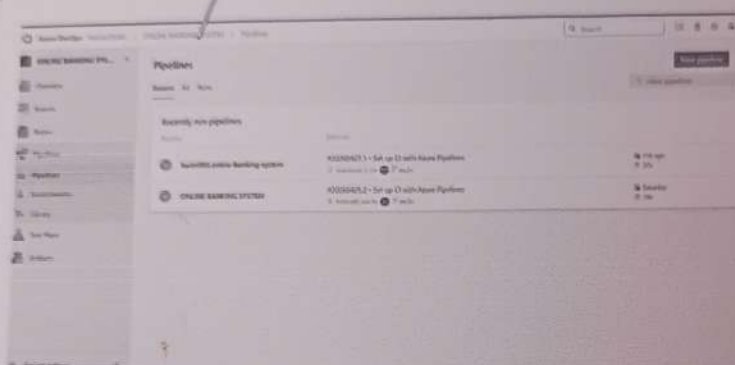
4. Running the test :

⇒ Executed the load test

⇒ Monitored system performance

5) Analyzing Results :

⇒ Analyzed performance using graphs and reports



Pipeline Setup :

⇒ Connecting Github to Azure DevOps

⇒ Created a project in Azure DevOps

⇒ Connected the Github repository

⇒ Creating pipeline :

⇒ Created a new pipeline

⇒ Selected Github as the source

3) Creating Yaml file:

⇒ Created azure-pipelines.yml file in the repository

⇒ Defined the pipeline configuration

4) Pipeline Configuration Steps:

⇒ checkout source code

⇒ setup python environment

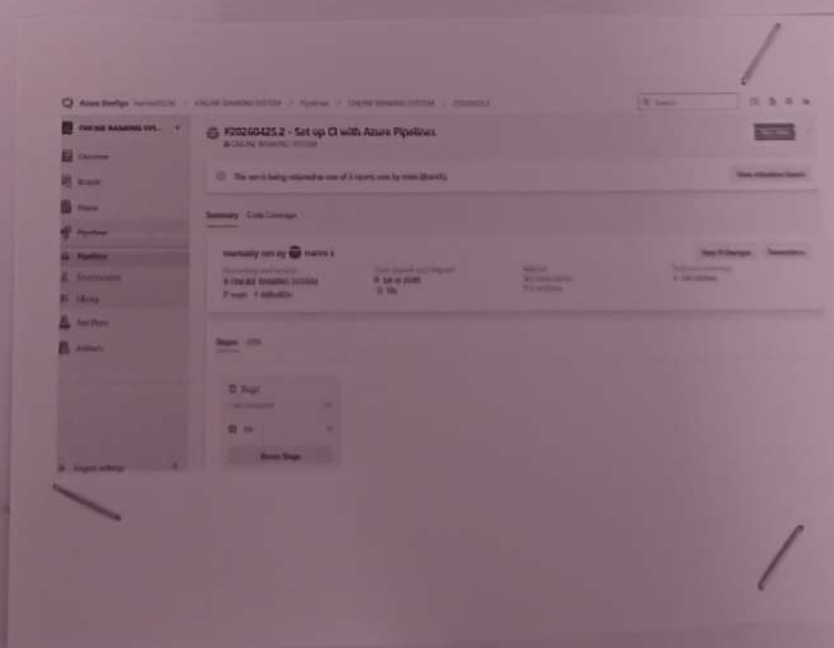
5) Triggering the pipeline

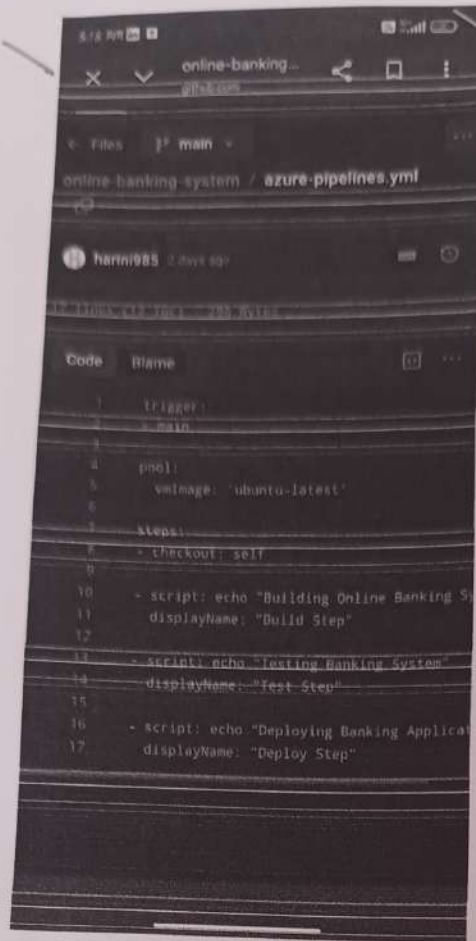
⇒ pipeline was automatically triggered on every code commit.

6) Monitoring pipeline:

⇒ Verified execution using logs in Azure DevOps.

⇒ checked for 'successful' build & errors.





The screenshot shows a mobile interface for an Azure Pipelines workflow named 'online-banking-system / azure-pipelines.yml'. The workflow is authored by 'harni985' and was last updated '2 days ago'. It features a table with columns 'Code' and 'Name' listing the steps of the pipeline:

Code	Name
1	trigger: main
2	pool: vmImage: 'ubuntu-latest'
3	steps:
4	- checkout: self
5	- script: echo "Building Online Banking System"
6	displayName: "Build Step"
7	- script: echo "Testing Banking System"
8	displayName: "Test Step"
9	- script: echo "Deploying Banking Application"
10	displayName: "Deploy Step"

Ex: 10:

Arm:

View of
checking
early
phase

Result:

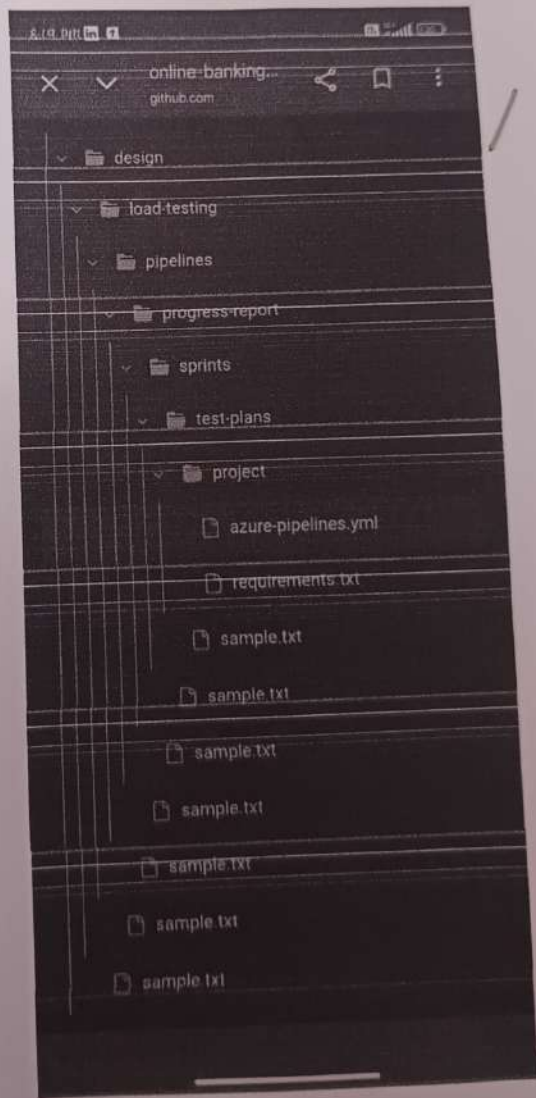
The load testing setup helped in
evaluating the performance of the online
Banking System

Ex: 10:

GitHub : Perfect Structure & Naming Conventions :

Aim:

To provide a clear and organized view of the project folder structure and file naming conventions, helping contributors and users easily understand, navigate and extend the Music playlist Batch Creator project.



Result:

The Result: displays the organized project structure and making it easy for users and contributors to understand and contribute to the codebase.